TAO Wenwen

Targeted Academic CV — Adaptive Neuromodulation

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Research Profile

Clinically trained neuroscience researcher interested in how measured brain and behavioral states can guide individualized neuromodulation. My experience spans depression rTMS, MRI-guided bilateral GPi transcranial ultrasound stimulation, repeated symptom and cognitive monitoring, multimodal MRI, EEG–fNIRS and physiological sensing, providing a foundation for state-dependent and closed-loop neurotechnology.

Research Interests

Adaptive Neuromodulation · State-Dependent Stimulation · rTMS · Transcranial Ultrasound Stimulation · MRI-Guided Targeting · Neural-State Measurement · Closed-Loop Neurotechnology

Education

Nanchang University | B.M. in Clinical Medicine | 2017–2022

GPA: 3.2/4.0 (82/100). Relevant training: Psychiatry, Neurology, Internal Medicine, Pathology, Pharmacology, Medical Statistics and Evidence-Based Medicine.

Publications and Manuscripts

1. **Tao, W.** Early Depressive Symptom Severity and Six-Month Return to Drinking After Inpatient Treatment for Alcohol Dependence: A Prospective Cohort Study. Manuscript submitted to *Journal of Substance Use and Addiction Treatment*.
2. **Tao, W.** Temporal Dataset Shift Limits Relapse-Risk Stratification Across the Inpatient-to-Community Transition in Alcohol Dependence: A Temporal Comparison of Conventional Machine Learning and Self-Supervised Representation Learning. Manuscript submitted to *Addiction Science & Clinical Practice*.
3. **Tao, W.** Task state dominates participant-held-out prediction of risky decisions in methamphetamine dependence, with limited incremental value from pre-decision EEG. Manuscript submitted to *Clinical EEG and Neuroscience*.

Research Experience

Tsinghua University — Yan Lab | Sep 2025–Mar 2026

Depression Neuroimaging and Neuromodulation Research

Research Intern | Advisor: Prof. Chaogan Yan

Project: *Mind Flower — rTMS Neuromodulation and Mindfulness-Based Counseling for Depression*

Multi-phase support: NSFC, National Key R&D Program, STI2030 Brain Science and CAS

- Investigated longitudinal clinical and multimodal markers of depression and rTMS treatment response by screening participants and administering MINI, HAMD, HAMA and MADRS assessments at baseline, intervention and follow-up.
- Managed resting-state, rumination and distraction fMRI acquisition, including individualized negative-recall cues, participant-specific E-Prime substitution, real-time head-motion monitoring (<3 mm) and OptiMRI think-aloud audio quality control.
- Tracked repeated rTMS sessions, daily pre-treatment HAMD-6 variation and adverse effects, creating structured dose–response and tolerability records alongside WCST, SST and emotional n-back assessments.
- Integrated laboratory fMRI and behavioral phenotyping with two-week wearable monitoring of HRV, sleep and activity and longer-term clinical follow-up.
- Coordinated clinicians, counselors and participants across multi-month cycles and standardized E-Prime recovery, export, multimodal naming, QC and access-controlled NAS archiving.

Beijing Normal University — Liu Lab | Sep 2025–Mar 2026

Parkinson's Disease Neuromodulation Research

Visiting Research Intern | Advisor: Dr. Zheyun Liu

Project: *MRI-Guided Transcranial Ultrasound Stimulation of Bilateral GPi for Cognitive Modulation in Parkinson's Disease*

- Supported full-protocol transcranial ultrasound stimulation targeting the bilateral globus pallidus internus using structural MRI and Brainsight neuronavigation for individualized deep-target localization.
- Documented stimulation parameters, including pulse-repetition frequency and duty cycle, and maintained target, session and safety records for reproducible protocol execution.
- Operated a combined TUS–cognitive protocol with approximately 1.5 hours of memory and cognitive-flexibility assessment and synchronized behavioral-response recording.
- Monitored participant-reported auditory, thermal and other tolerability signals; the study evaluates whether non-invasive deep-brain stimulation can modulate basal-ganglia-related cognition under standardized safety procedures.

Tsinghua University — CogDrive Lab | Mar 2026–Present

State Key Laboratory of Intelligent Green Vehicle and Mobility

Research Assistant | Advisor: Prof. Dongpu Cao

Project: *Multimodal AI for Adolescent Depression Screening and Human–AI Interaction in Intelligent Cockpits*

Clinical collaborator: Tongji Hospital, Huazhong University of Science and Technology

- Independently conceived and led an IRB-approved prospective multimodal cohort protocol for children and adolescents aged 6–18, with planned enrollment of 300 clinical participants and 150 healthy controls across four developmental bands; real-world recruitment and acquisition remain in the pre-collection hardware-validation stage.
- Established a four-stage clinical ground-truth framework—CDRS-R, age-matched self/parent reports, K-SADS-PL DSM-5 clinician confirmation and multisource severity grading—and mapped item-level clinical anchors to facial, speech, gaze and physiological features.
- Designed synchronized acquisition across 13+ streams, including simultaneous 64-channel EEG–fNIRS, desktop eye tracking, multi-view facial video, speech, ECG, PPG, GSR and respiration during computerized paradigms and clinician-led interviews.
- Specified event and trigger conventions, tiered temporal alignment, metadata, operator procedures, signal-quality labels and analysis-ready schemas for multimodal state modeling and exploratory EEG–fNIRS neurovascular-coupling analysis.

Shenzhen Mental Health Center — Department of Addiction Medicine | Jan 2024–Aug 2025

Addiction Neuroscience and Translational Research

Research Assistant, Full-time | Advisors: Prof. Mei Yang and Prof. Xiaojian Jia

Project: *Longitudinal Addiction Cohorts, Multimodal Neuroimaging and Gut–Brain Research*

- Managed research workflows for 290+ participants with alcohol use disorder, methamphetamine dependence and matched controls, integrating longitudinal clinical phenotypes, MRI/DTI, EEG, cognition and biospecimens in a standardized database with >95% completeness.
- Investigated early depressive symptoms and six-month return to drinking in 107 inpatients using prospective follow-up and covariate-adjusted models; the work is reported in a manuscript submitted to the Journal of Substance Use and Addiction Treatment.
- Evaluated temporal dataset shift in relapse-risk stratification across inpatient and community phases, comparing conventional machine learning with self-supervised representations and emphasizing transportability rather than in-sample performance.
- Analyzed participant-held-out risky-decision prediction in methamphetamine dependence, finding that task state dominated prediction and that pre-decision EEG provided limited incremental value; the manuscript is submitted to Clinical EEG and Neuroscience.

Clinical Experience

Wannian County Hospital of TCM | Resident Physician | Dec 2022–Dec 2023

- Managed 15–20 chronic-disease inpatients through assessment, treatment planning, inpatient care and follow-up and developed standardized medical-record templates for longitudinal traceability.

Ganzhou People's Hospital | Clinical Intern | Feb 2021–Feb 2022

- Completed 12 clinical rotations, including Neurology ICU and Cardiology ICU, and performed supervised procedures including lumbar puncture and ECG interpretation.

Methods and Technical Expertise

Clinical and Psychiatric Assessment: MINI; K-SADS-PL DSM-5; CDRS-R; HAMD; HAMA; MADRS; PHQ-9; PHQ-15; CDI; SDQ; longitudinal cohort design; IRB coordination; consent/assent; pediatric safeguards

Neuroimaging: Structural MRI; resting-state and task fMRI; DTI tractography; FA/ADC; structural–functional connectivity; FSL; DPABI; DPABI-Surf; MRI-based targeting

Neurophysiology and Multimodal Sensing: 32/64-channel EEG; simultaneous EEG–fNIRS; ECG; PPG; GSR; respiration; desktop eye tracking; synchronized video and speech; event-trigger design; signal QC

Neuromodulation: rTMS; transcranial ultrasound stimulation; Brainsight neuronavigation; individualized targeting; stimulation-parameter logging; tolerability and safety monitoring

Behavioral and Cognitive Paradigms: MMN; VFT; N-back; emotional n-back; WCST; SST; BART; two-step task; E-Prime and PsychoPy workflows

Programming and Quantitative Methods: Python; R; SPSS; MATLAB; logistic regression; longitudinal modeling; participant-held-out validation; temporal transportability; multimodal data schemas

Research Infrastructure and Data Engineering: BPlus MVP-0; synchronization and trigger architecture; metadata and automated QC; longitudinal databases; data dictionaries; version control; NAS archiving; multicenter harmonization

Languages

Chinese: Native | English: IELTS 6.5; TOEFL 97